Exercise Solution for Chapter-02: Wireless Transmission

1. **Frequency regulations may differ between countries. Check out the regulations valid for your country (within Europe the European Radio Office may be able to help you, www.ero.dk, for the US try the FCC, www.fcc.gov, for Japan ARIB, www.arib.or.jp).**

For Bangladesh, frequency regulation is managed by the Bangladesh Telecommunication Regulatory Commission (BTRC). They control spectrum allocation, licensing, and equipment approval. Common ISM bands like 2.4 GHz and 5 GHz are allowed for unlicensed use. Importing and operating radio equipment requires prior approval and licensing from BTRC.

2. **Why can waves with a very low frequency follow the earth's surface? Why are they not used for data transmission in computer networks?**

Radio waves with frequencies below 2 MHz can follow the earth's surface due to **ground wave propagation**. There are two main reasons behind this:

1. **Diffraction**: Low-frequency waves have long wavelengths, which allows them to bend around large obstacles such as hills and buildings. This helps them to "hug" the curvature of the Earth.
2. **Surface current induction**: The wave induces a current in the Earth's surface, which slows the portion of the wave-front near the ground. This causes the wave-front to tilt downward and follow the Earth's contour.

However, these low frequencies are generally **not used for data transmission in computer networks** for several key reasons:

- **Low data rates**: According to **Nyquist and Shannon's theorems**, lower frequencies offer lower bandwidth, which limits the maximum achievable data rate.
- **Large antennas required**: Efficient transmission and reception at low frequencies require **very large antennas** (often impractical for consumer devices like mobile phones or Wi-Fi routers).
- **Difficult spatial reuse**: Low-frequency signals **penetrate materials more easily**, making **Spatial Division Multiplexing (SDM)** harder. This would result in **large cell sizes** and **poor frequency reuse**, which is unsuitable for dense, high-speed networks.

3. **Why does the ITU-R only regulate 'lower' frequencies (up to some hundred GHz) and not higher frequencies (in the THz range)?**

The **International Telecommunication Union – Radiocommunication Sector (ITU-R)** primarily regulates lower frequencies (up to several hundred GHz) because:

- **THz frequencies** (e.g., infrared, visible light) are **easily blocked by obstacles** and do **not interfere** with other transmissions, making regulation less critical.
- At such high frequencies, **standard safety regulations** (like those for **laser emissions**) are typically sufficient.
- Most practical **radio systems operate below 100 GHz** because:
  - It is **technically challenging** to generate and handle higher frequencies in the **THz range**.
  - Equipment for THz frequencies is **complex and expensive**, limiting widespread use.

Therefore, due to limited interference potential and technical limitations, the ITU-R focuses its regulation on the more commonly used, interference-prone **lower frequency bands**.

4. What are the two different approaches in regulation regarding mobile phone systems in Europe and the US? What are the consequences?

1. **European Approach:**
   - Based on standardization and regulation before product availability.
   - Governments founded ETSI (European Telecommunications Standards Institute) to harmonize national regulations.
   - ETSI developed standards that **all EU countries were required to follow**.

2. **US Approach:**

   - Driven by companies and market forces.
   - Companies **develop systems first**, and **try to standardize** them later.
   - The **FCC (Federal Communications Commission)** focuses on **fairness** between competing systems rather than mandating a specific one.

   Consequences of the Two Approaches:

Aspect	European Approach	US Approach
Basis	Standards before products	Products before standards
Regulation	ETSI sets mandatory standards	FCC ensures fairness only
Outcome	Some failures, some European-only, some global hits (GSM)	Fast product success (Wi-Fi), but fragmented mobile systems
Features	Many advanced features (roaming, MMS)	Some features missing or delayed

5. Why is the international availability of the same ISM bands important?

Unlike stationary devices like TVs, computers—such as laptops and PDAs—are used globally. Users expect their built-in WLAN adapters to work anywhere without the need for reconfiguration or licensing. For manufacturers, creating a single, standardized system is much more cost-effective than designing multiple versions for different regions with smaller markets.

6. Is it possible to transmit a digital signal, e.g., coded as square wave as used inside a computer, using radio transmission without any loss? Why?

No, **loss-free transmission of analog or digital signals is not possible** due to factors like **attenuation** and **dispersion** that distort the signal during transmission.

- ✓ A **digital signal**, such as a square wave, is actually composed of a **bundle of analog sine waves** (per Fourier analysis).
- ✓ To perfectly represent a square wave, an **infinite number of sine waves** at different frequencies are required.
- ✓ However, **no transmission medium can carry infinitely high frequencies**.
- ✓ Therefore, the digital signal **cannot be transmitted without any loss or distortion**.

7. Is a directional antenna useful for mobile phones? Why? How can the gain of an antenna be improved?

Without any additional intelligence, directional antennas are **not useful in standard mobile phones** because users frequently move, rotate, or flip their devices and often operate them hands-free or while the phone is in a pocket or bag. This makes directed transmission impractical. However, recent advancements using fast signal processors and multiple antennas enable beamforming, which can utilize directional antenna characteristics effectively. **Antenna gain** can be improved through correct dimensioning (e.g., antenna length equal to half the wavelength), using multiple antennas with signal processing, or adding active and passive components—similar to those used in traditional TV antennas or satellite dishes.

8. **What are the main problems of signal propagation? Why do radio waves not always follow a straight line? Why is reflection both useful and harmful?**

The main problems of signal propagation include **attenuation**, **scattering**, **diffraction**, **reflection**, and **refraction**. Except for attenuation, all these effects can cause radio waves to **deviate from a straight path**. In reality, radio waves **only travel in straight lines in a perfect vacuum without gravitational influences**. In urban environments, reflection is particularly important—**without it, radio reception in cities would often be impossible**, since a clear line-of-sight rarely exists. However, **reflection also leads to multipath propagation**, which can cause **Inter-Symbol Interference (ISI)**, making it both **useful and harmful**.

9. **Name several methods for ISI mitigation. How does ISI depend on the carrier frequency, symbol rate, and movement of sender/receiver? What are the influences of ISI on TDM schemes?**

ISI Mitigation Methods:

- Guard Spaces: Add time gaps between symbols to prevent overlap.
- Low Symbol Rate: Slower transmission reduces the chance of symbols interfering.
- OFDM: Spreads data across many carriers at low symbol rates to reduce ISI.
- Channel Estimation: Detect and track the strongest signal paths and adjust the receiver to match.

Dependence Factors:

- Higher Frequencies: Reduce multipath effects since waves behave more like light, lowering ISI.
- Higher Symbol Rates: Increase ISI because symbols are packed too closely in time.
- Mobility: Moving sender or receiver causes changing reflections, making ISI worse and harder to correct.

Impact on TDM: ISI reduces efficiency in TDM systems because longer guard times are needed between time slots, lowering the available bandwidth.

10. **What are the means to mitigate narrowband interference? What is the complexity of the different solutions?**

- **Dynamic Frequency Selection (DFS)** – Avoids interference by choosing a cleaner frequency. – **Low complexity** (easy to implement).
- **Frequency Hopping (FH)** – Switches between frequencies to avoid staying on a bad one. – **Medium complexity** (needs timing sync).
- **Direct Sequence Spread Spectrum (DSSS)** – Spreads the signal over a wide range using a code. – **High complexity** (needs strong receivers to decode).

11. **Why, typically, is digital modulation not enough for radio transmission? What are general goals for digital modulation? What are typical schemes?**

Why not enough? Digital signals must be **modulated onto analog carrier frequencies** to allow **multiple systems to share the radio spectrum** using FDM (Frequency Division Multiplexing). For example, many radio stations use the same base frequency range (e.g., voice), so they must be transmitted on different carriers.

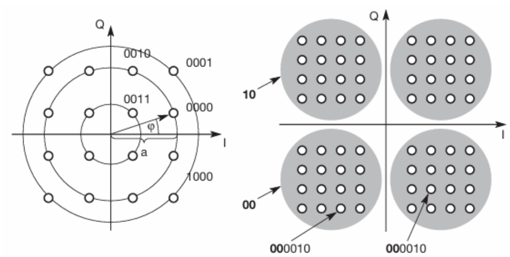
Goals of digital modulation:

- **Spectral efficiency** (use less bandwidth)
- **Power efficiency** (use less energy)
- **Robustness** (resist noise and interference)

Typical schemes:

- ASK (Amplitude Shift Keying)
- PSK (Phase Shift Keying)
- FSK (Frequency Shift Keying)

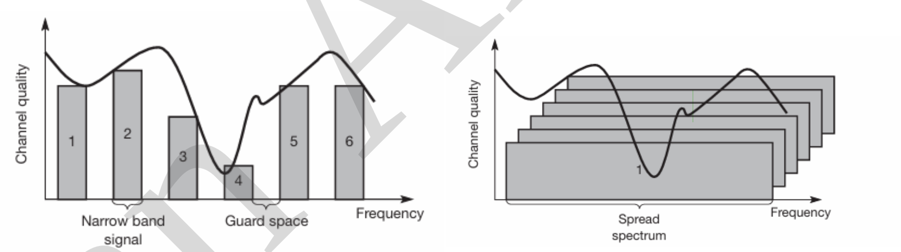
12. **Think of a phase diagram and the points representing bit patterns for a PSK scheme (see Figure below).**
    i. **How can a receiver decide which bit pattern was originally sent when a received 'point' lies somewhere in between other points in the diagram?**
    ii. **Why is it difficult to code more and more bits per phase shift?**



The receiver **compares the received point** with all possible points in the **constellation diagram** and picks the **closest one**. That point represents the original data.

It becomes difficult to **add more bits per phase shift** because more points mean **smaller gaps** between them. If **noise** shifts a point, it may be **misinterpreted as another**, causing errors. So, the more points (i.e., the higher the data rate), the **higher the risk of mistakes** due to noise.

13.
    i. **What are the main benefits of a spread spectrum system?**
    ii. **How can spreading be achieved?**
    iii. **What replaces the guard space in Figure 2.33 (left below) when compared to Figure 2.34 (right below)?**
    iv. **How can DSSS systems benefit from multi-path propagation?**



Main benefits:

- Strong protection against interference
- Better security (hard to intercept without the spreading code)
- Supports CDMA systems
- Can coexist with other systems if signal strength is low

Spreading methods:

- **XOR** with a chipping sequence (DSSS)
- **Frequency hopping**

Guard space replacement: Replaced by **orthogonal codes** (in DSSS) or **hopping patterns** (in FHSS) to avoid interference.

DSSS benefit from multipath: Uses **RAKE receivers** to combine signals from different paths, making the final signal **stronger and more reliable**

14. What are the main reasons for using cellular systems? How is SDM typically realized and combined with FDM? How does DCA influence the frequencies available in other cells?

Why cellular systems?
- ✓ Support more users by reusing frequencies across cells
- ✓ Smaller cells = more users/km², lower power, less radiation, longer battery life, and faster communication
- ✓ Enable location-based services

How SDM + FDM work:
- ✓ Space Division Multiplexing (SDM): divides the area into cells
- ✓ Frequency Division Multiplexing (FDM): assigns different frequencies to each cell

DCA (Dynamic Channel Allocation):
- ✓ Reacts to traffic by borrowing frequencies from nearby cells
- ✓ But borrowed frequencies **must be blocked** in neighboring cells to avoid interference

15. What limits the number of simultaneous users in a TDM/FDM system compared to a CDM system? What happens to the transmission quality of connections if the load gets higher in a cell, i.e., how does an additional user influence the other users in the cell?

- TDM/FDM has a hard limit – each user gets a fixed time slot and frequency. If all are used, no new users can join.
- CDM has a soft limit – it can accept more users, but each one adds noise, reducing signal quality for everyone.
- In TDM/FDM, users are isolated, so adding one doesn't affect others.
- In CDM, every new user makes the signal weaker for all users.